

Bio Informatics

Lecture 12

Scoring Matrices - Types

- **Identity** scoring - the simplest scoring scheme, where characters are classified as: identical (scores 1) , or non-identical (scores -1 or 0). This scoring scheme is not much used.
- **Chemical similarity** scoring (for proteins) - this matrix gives greater weight to amino acids with similar chemical properties (e.g. size, shape or charge of the amino acid).

Scoring Matrices - Types

- **Observed Frequency matrices** for proteins - most commonly used by all programs.
- These matrices are constructed by analyzing the substitution frequencies seen in the alignments of known families of proteins.

Observed Frequency Matrices

- Every possible identity and substitution is assigned a score.
- This score is based on the observed frequencies of such occurrences in alignments of evolutionary related proteins.
- This score will also reflect the frequency that a particular amino acid occurs in nature, as some amino acids are more abundant than others.

Observed Frequency Matrices

- Identities are assigned the most positive scores.
- Frequently observed substitutions also receive positive scores.
- Mismatches, or matches that are unlikely to have been a result of evolution, are given negative scores.

Observed Frequency Matrices

- Each matrix entry gives the ratio of the observed frequency of substitution between each possible pair of amino acids in related proteins to that expected by chance, given the frequencies of amino acids in proteins.
- These ratios are called odds scores.
- These ratios are transformed to logarithms of odds scores called log odds scores.
- Odds scores and log odds scores are used to score protein alignments

Examples of Ubiquitin

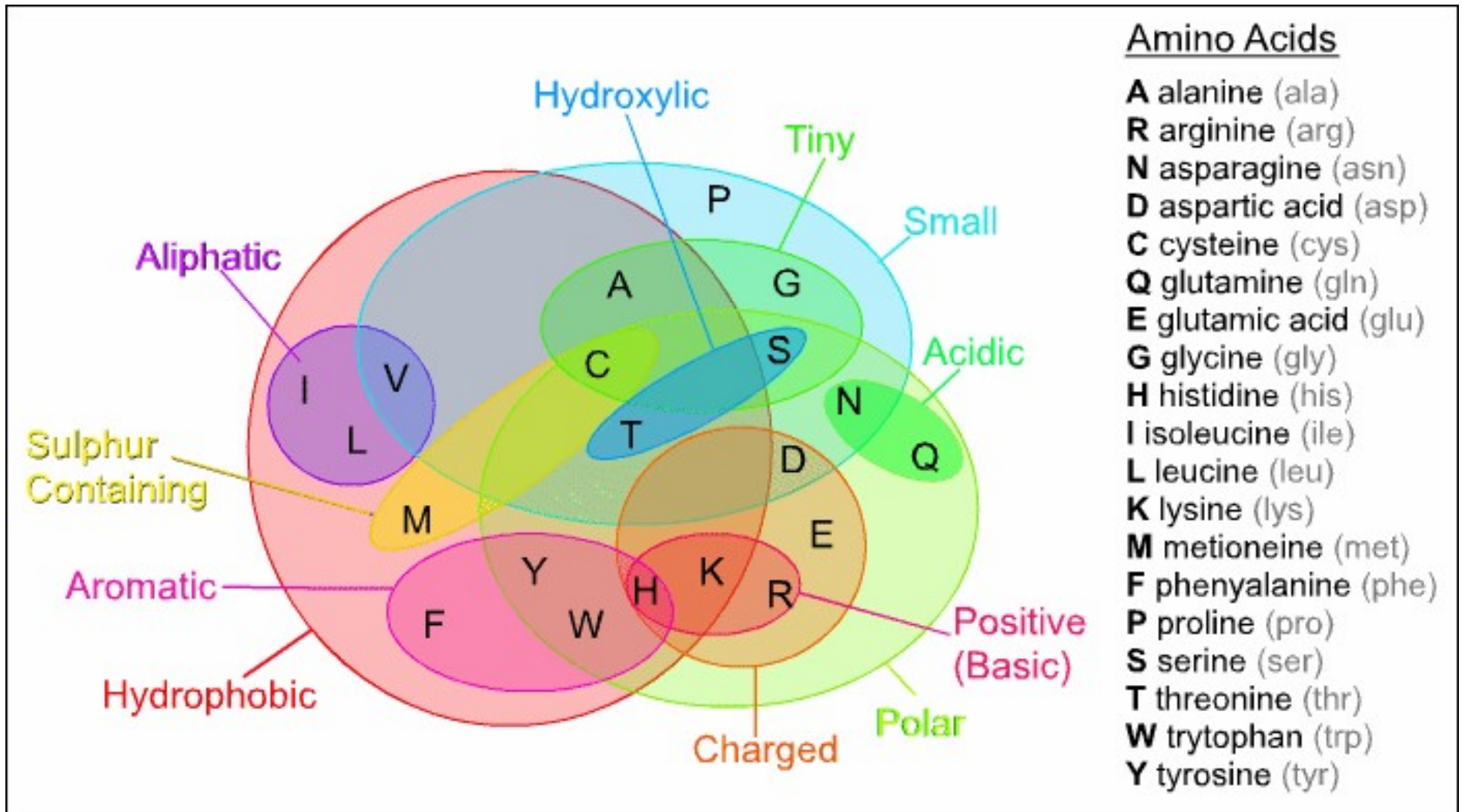
Human	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	T	S	G	T	S	P	V	H
Chimp	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	T	S	G	T	S	P	V	H
Orangutan	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	T	S	G	T	S	P	V	H
Macaque	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	T	S	G	T	S	P	V	H
Mouse	V	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	A	S	G	T	S	P	V	H
GuineaPig	V	A	Q	E	L	L	K	Q	G	A	S	P	N	V	Q	D	A	S	G	T	S	P	L	H
Dog	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	A	T	G	T	T	P	A	H
Horse	I	A	L	E	L	L	K	Q	G	A	N	P	N	I	Q	D	A	S	G	T	T	P	A	H
Cow	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	A	S	G	T	T	P	A	H
Opossum	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	G	S	G	T	S	P	A	H
Platypus	I	A	L	E	L	L	K	Q	G	A	N	P	N	V	Q	D	A	T	G	T	T	P	V	H
Frog	V	A	S	E	L	L	K	Q	G	A	S	P	N	I	Q	D	A	Y	G	T	T	P	A	H
Stickleback	I	A	G	L	L	L	E	K	G	A	D	P	N	V	Q	D	N	H	G	I	A	P	V	H
Fugu	I	A	S	L	L	L	E	K	G	A	D	P	N	V	Q	D	K	H	G	I	A	P	V	H
Tetraodon	I	A	S	L	L	L	E	K	G	A	D	P	N	V	Q	D	K	H	G	I	A	P	V	H
Medaka	I	A	S	L	L	L	E	K	G	A	D	A	N	V	Q	D	K	H	G	I	A	P	V	H

Examples of Ubiquitin

Human	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	T	S	G	T	S	P	V	H
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Tetraodon	I	A	S	L	L	L	E	K	G	A	D	P	N	V	Q	D	K	H	G	I	A	P	V	H
Medaka	I	A	S	L	L	L	E	K	G	A	D	A	N	V	Q	D	K	H	G	I	A	P	V	H

$$s(a, b) = \frac{1}{\lambda} \log \frac{P_{ab}}{f_a f_b}$$

Chemical properties of Amino Acids



Occurrence frequencies of 20 AAcids

The average amino acid frequencies reported by Robinson and Robinson in [173].

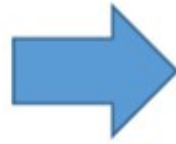
Amino acid	Frequency	Amino acid	Frequency	Amino acid	Frequency	Amino acid	Frequency
Ala	0.078	Gln	0.043	Leu	0.090	Ser	0.071
Arg	0.051	Glu	0.063	Lys	0.057	Thr	0.058
Asn	0.045	Gly	0.074	Met	0.022	Trp	0.013
Asp	0.054	His	0.022	Phe	0.039	Tyr	0.032
Cys	0.019	Ile	0.051	Pro	0.052	Val	0.064

Example of deriving scoring matrix

- Lets try to compute a scoring matrix from the following series of 6 sequences of comprising of 3 Amino acids A, B and C.
- B A B A
- A A C A
- A A C C
- A A B A
- A A B C
- A A B B

Example of deriving scoring matrix

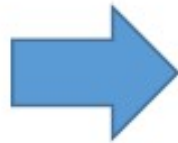
B A B A
A A C A
A A C C
A A B A
A A B C
A A B B



<i>aligned pairs</i>	<i>observed frequencies f_{ij}</i>
A to A	28/60
A to B	8 /60
A to C	6/60
B to B	6/60
B to C	10/60
C to C	2/60

Example of deriving scoring matrix

B A B A
A A C A
A A C C
A A B A
A A B C
A A B B



<i>aligned pairs</i>	<i>observed frequencies f_{ij}</i>
A to A	28/60
A to B	8 /60
A to C	6/60
B to B	6/60
B to C	10/60
C to C	2/60



<i>amino acids</i>	<i>observed frequencies f_i</i>
A	14/24
B	6/24
C	4/24

Example of deriving scoring matrix

(i, j) -entry of matrix is $s_{ij} = \log_2 \frac{f_{ij}}{p_{ij}}$, $p_{ij} = \begin{cases} f_i f_i & i = j \\ 2f_i f_j & i \neq j. \end{cases}$

<i>aligned pairs</i>	<i>observed frequencies f_{ij}</i>	<i>expected frequencies p_{ij}</i>	$2s_{ij}$
A to A	28/60	196/576	0.91
A to B	8/60	168/576	-2.26
A to C	6/60	112/576	-1.92
B to B	6/60	36/576	1.35
B to C	10/60	48/576	2.00
C to C	2/60	16/576	0.52

<i>amino acids</i>	<i>observed frequencies f_i</i>
A	14/24
B	6/24
C	4/24

	A	B	C
A	1	-2	-2
B		1	2
C			1

Observed Frequency Matrices

- Observed Scoring Matrices are superior to simple identity scores, or scores based solely on chemical properties of amino acids.
- The most frequently used observed log odds matrices used are the PAM and BLOSUM matrices. So we will discuss them in next lectures.